

# Specialization in a Knowledge Economy

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This paper documents two trends in 1980s and 1990s:

1. US firms narrowed their scope of production.
  - Shrinkage mostly driven by innovative firms.
2. Innovation shifted from large firms to small firms.

Higher degree of separation between innovation and production.

## **Question:**

"What are the driving forces of the observed specialization, and how do they affect economic growth?"

## Hypothesis:

Pro-patent reforms in the 1980s and 1990s.

- Increased the trading of innovation between firms.

## Model

Firms heterogeneous w.r.t productivity and innovation level.

Firms choose production scope and innovation effort.

Innovation might fall outside a firm's scope.

- Firms buy and sell innovation in a market with search frictions.

More efficient market + more protection to sellers  $\Rightarrow$

- Less incentive to increase scope.
- More incentive for small firms to innovate.

## Estimation Results:

Increasing in market efficiency + more protection to sellers:

- Accounts for 30% of scope decrease.
- Accounts for 49% of reallocation of innovation activities.
- Increase growth rate in 0.68%

## Empirical Analysis:

Uses regional and firm-level differences in the exposure to patent policy changes.

- Evidence of causality from patent reforms to decrease in production scope of firms.

# Specialization Patterns

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# Measurements

Longitudinal Business Database (LBD)

- **Scope:** Aggregate the NAICS codes of every establishment under a particular firm's ID.

Patent Data Project<sup>1</sup> (PDP).

- **Innovating firm:** Has been granted a patent between 1976 and 2006.

Survey of Industrial Research and Development (SIRD).

- **R&D intensity:** ratio of R&D cost to net domestic sales.

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<sup>1</sup>Collected by the NBER

# Production Scope

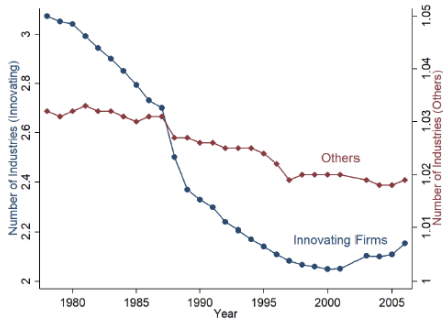
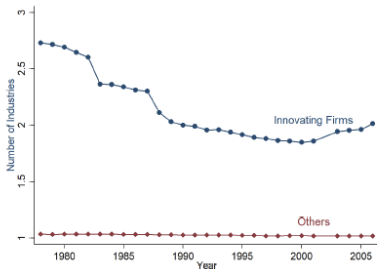


Figure 1: Trend of Production Scope by Innovating Activities

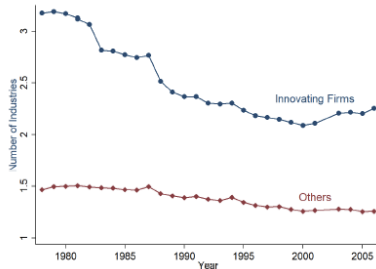
# Production Scope

Regression (For each year):

Scope : innovating dummy + employment + innovating x employment



(a) 20 Employees

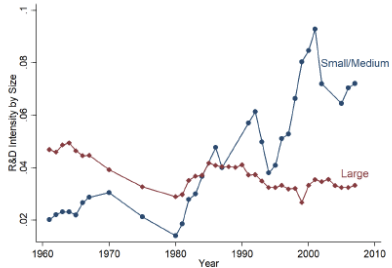


(b) 1000 Employees

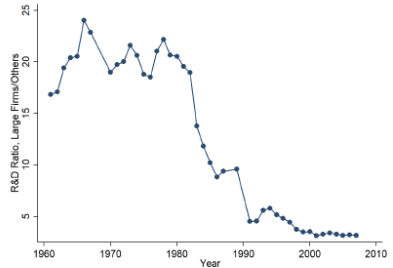
Figure 2: Trends of Production Scope with Fixed Firm Size

# Trends of R&D activity

**R&D intensity:** ratio of R&D cost to net domestic sales. Large firms: 1000+ employees.



(a) R&D Intensity by Size



(b) Total R&D Expense Ratio

Figure 3: Trends of R&D Activities

# Policy Reforms and Patent Market

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1980s: Policies to stimulate innovation and boost growth.

## 1981: Extension of Patentability to Softwares and Genetic Engineering

- Patent issuance and application both doubled afterwards.

1981: Tax credit on R&D

1982: Creation of the Court of Appeals for the Federal Circuit.

1982: Longer Patent Terms for Most Inventions.

# Event: Court of Appeals for the Federal Circuit.

Post CAFC - Higher success rate of patent holders in legal disputes.

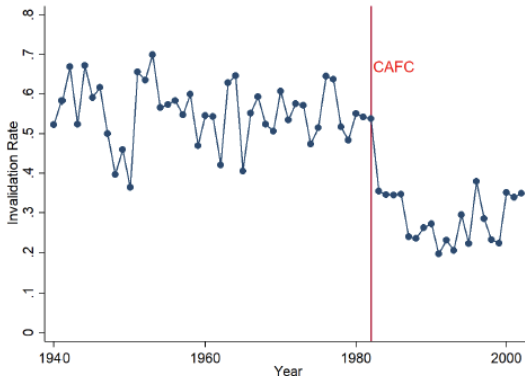


Figure 4: Fraction of Lawsuits Invalidating the Patents

# Increased tradability of innovation.

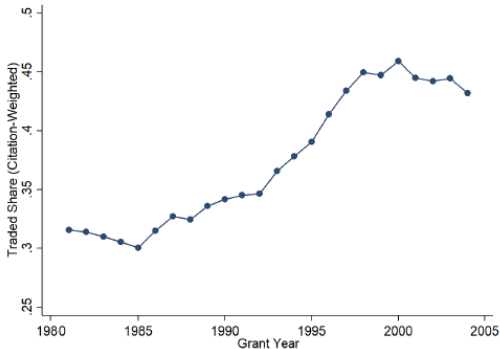


Figure 5: Fraction of Patents Involved in Trading

↑ Incentives to patenting  $\Rightarrow$  ↑ Information disclosure  $\Rightarrow$  ↑ Trading efficiency.

Balanced growth path model with heterogeneous firms.

Allows to estimate the importance of patent market efficiency vis a vis other possible explanations for the patterns:

- Increase in R&D subsidy.
- Change in cost structure of increasing scope. - Hshie and Rossi-Hansberg (2019)
- Good ideas getting harder to find. - Bloom et al. (2019)

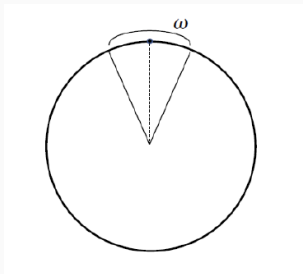
# Model

There is unit mass of firms located around a industry circle.

Firms start the period with a production ability,  $m \in \{m_H, m_L\}$ , and an innovation level,  $z$ .

- $m$  follows Markov Process:  $Q'_{mm}$

**First sub-period:** Choose production scope,  $\omega$ , and innovation probability,  $i$  - R&D effort.



**Second sub-period:** Firms Innovation level is updated:

$$z' = z + \gamma(\mathbb{1}_{(RD \in \omega)} + \mathbb{B})Z$$

$\mathbb{1}_{(RD \in \omega)} = 1$  with prob.  $iX(|\omega|)$ ,  $X'(|\omega|) > 0$ ,  $X''(|\omega|) < 0$

$\mathbb{B}$  depends on the matching process

$Z$  is the productivity-weighted average innovation level.

Cost of increasing  $\omega$  and  $i$ :

$$C^e(\omega; Z) = \mu \frac{|\omega|^{1+\iota}}{1+\iota} Z^{\zeta/(\zeta+\lambda)}, \quad \iota > 0$$

$$C^i(i; Z) = \chi \frac{i^{1+\rho}}{1+\rho} Z^{\zeta/(\zeta+\lambda)}, \quad \rho > 0$$

Finally, firms choose  $k$  and  $l$  for production:

$$\pi(\omega, m, z'; Z) = \max_{k,l} (mz')^{\zeta} (|\omega|k)^{\eta} (|\omega|l)^{\lambda} - \tilde{r}(|\omega|k) - w(|\omega|l)$$

# Model - Patent market

Number of matches in an arc  $d$ :

$$M(\eta_s(d), n_b(d)) = \phi n_s(d)^\eta n_b(d)^{1-\eta}$$

$n_s(d), n_b(d)$ : Search effort by sellers and buyers on arc  $d$ .

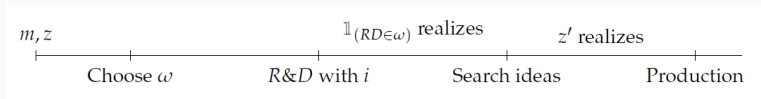
In a symmetric equilibrium:

$$b(\omega) = \phi \left( \frac{\eta_s}{\eta_b} \right)^\eta, \quad s = \phi \left( \frac{\eta_b}{\eta_s} \right)^{1-\eta}$$

$$p_b(m, z; \mathbf{Z}) = \arg \max_{p_b} p_b^\theta [\pi(m, z'; \mathbf{Z}) + EV(m', z'; \mathbf{Z}) - p_b - (\pi(m, z; \mathbf{Z}) + rEV(m', z, \mathbf{Z}'))]$$

Seller's expected price:  $p_s(\mathbf{Z}) = \int \int p_b(m, z, \mathbf{Z}) dG(m, z; \mathbf{Z})$

# Model - Value functions



$$\begin{aligned}
 D(\omega, m, z, Z) = & \max_i \{ iX(|\omega|) \underbrace{[\pi(m, z', Z) + rEV(m', z'; Z')]}_{\text{Innovate withing } \omega} + \\
 & (1 - iX(\omega)) \underbrace{[b(\omega)(\pi(m, z'; Z) - p_b(m, z; Z') + rEV(m', z'; Z))]}_{\text{Buy an idea within } \omega} + \\
 & (1 - iX(\omega)) \underbrace{[(1 - b(\omega))(\pi(m', z; Z) + rEV(m', z; Z'))]}_{\text{no idea within } \omega} + \\
 & i(1 - X(\omega)) \underbrace{sp_s}_{\text{Sell and idea}} - (1 - \sigma)C^i(i; Z) \}
 \end{aligned}$$

with

$$V(m, z; Z) = \max_{\omega} D(\omega, m, z; Z) - C^e(\omega, Z)$$

$\sigma$ : R&D tax subsidy

## Inspecting the Mechanism

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# Impact of Change in Parameters

Increase in  $\phi$ :

$$i^*(\omega, m) = \left\{ \frac{\gamma}{(1-\sigma)\chi} [X(|\omega|) \underbrace{(1 - (1-\theta)b)B(m)}_{-} + \underbrace{(1 - X(|\omega|))s\theta(\sigma_H B(m_H) + \sigma_L B(m_L))}_{+}] \right\}^{\frac{1}{\rho}}$$

$$X'(|\omega|) \underbrace{i^*(\omega, m)}_{+/-} \underbrace{[(1 - (1-\theta)b)B(m) - s\theta(\sigma_H B(m_H) + \sigma_L B(m_L))]}_{-} \gamma = \mu |\omega|^\nu$$

Increase in  $\theta$ :

$$i^*(\omega, m) = \left\{ \frac{\gamma}{(1-\sigma)\chi} [X(|\omega|) \underbrace{(1 - (1-\theta)b)B(m)}_{+} + \underbrace{(1 - X(|\omega|))s\theta(\sigma_H B(m_H) + \sigma_L B(m_L))}_{+}] \right\}^{\frac{1}{\rho}}$$

$$X'(|\omega|) \underbrace{i^*(\omega, m)}_{+} \underbrace{[(1 - (1-\theta)b)B(m) - s\theta(\sigma_H B(m_H) + \sigma_L B(m_L))]}_{+/-} \gamma = \mu |\omega|^\nu$$

# Quantitative Analysis

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Evaluate four possible explanations for specialization:

- Changes in patent market:  $\phi$  and  $\theta$
- Increase in tax credits:  $\sigma$
- Changes in production cost structure:  $\mu$  and  $\iota$
- scarcity of goods ideas:  $\gamma$

Procedure:

- Set all the parameters to fit moments from 1981-1985.
- Change the parameters of interest, so the model fits 1996-2000.
- Evaluate the effect of changing one set of parameters at a time.

# Initial Calibration

Parameters from prior information:

| Parameter  | Description         | Value | Identification        |
|------------|---------------------|-------|-----------------------|
| $\eta$     | Capital share       | 0.25  | Guner et al. (2008)   |
| $\lambda$  | Labor share         | 0.60  | Guner et al. (2008)   |
| $\epsilon$ | CRRA parameter      | 2.00  | Standard              |
| $\beta$    | Discount factor     | 0.99  | Interest Rate         |
| $\delta$   | Depreciation rate   | 0.07  | NIPA                  |
| $\alpha_H$ | Share of high type  | 0.10  | Imposed               |
| $\alpha_L$ | Share of low type   | 0.90  | Imposed               |
| $\chi$     | R&D cost, scale     | 1.00  | Normalization         |
| $\sigma$   | R&D tax credit rate | 0.05  | Akcigit et al. (2018) |

Table 1: Parameter Values from a Priori Information

9.1% of innovating firms between 1981 and 2000, are large firms.

Only 5.5% of small firms have high productive ability

# Initial Calibration

Direct estimation:

$$m : \ln(emp_{ijt}) = \beta_1 \underbrace{\ln(patentStock_{ijt})}_{\ln(z')} + \beta_0 + u_t + v_j + \underbrace{residual_{ijt}}_{\ln(m)}$$

$$\nu : \ln(\#match_{jt}) = \alpha_0 + \nu \ln(\#seller_{jt}) + (1 - \nu) \ln(\#buyer_{jt}) + u_t + v_j + e_{jt}$$

Assume  $X(|\omega|) = \xi \omega^\psi$

$$\xi, \psi : \ln(WhithinScopeProb.) = \ln(\xi) + \psi \ln(NumerInd.) + \varepsilon_{jt}$$

| Parameter   | Description                 | Value                                                          | Identification |
|-------------|-----------------------------|----------------------------------------------------------------|----------------|
| $m_H$       | Prod. ability of high type  | 24.43                                                          | Regression     |
| $m_L$       | Prod. ability of low type   | 0.70                                                           | Regression     |
| $Q_{mm'}$   | Type transition matrix      | $\begin{bmatrix} 0.872 & 0.128 \\ 0.017 & 0.983 \end{bmatrix}$ | MLE            |
| $\nu$       | Matching function, exponent | 0.70                                                           | Regression     |
| $X(\omega)$ | Within-scope Probability    | $e^{-4.443} *  \omega ^{0.7643}$                               | Regression     |

Table 2: Parameter Values from Direct Estimations

# Initial Calibration

Parameters from GMM (targeting 1981-1985):

| Parameter   | Description                 | Value  | Identification                      |
|-------------|-----------------------------|--------|-------------------------------------|
| $\gamma$    | Lock-step of tech. growth   | 1.72   | Growth rate                         |
| $1 + \rho$  | R&D cost elasticity         | 1.79   | R&D cost/sales ratio (H and L)      |
| $\theta$    | Bargaining power            | 0.16   |                                     |
| $\mu$       | Management cost, scale      | 1.5E-4 | Avg. number of industries (H and L) |
| $1 + \iota$ | Management cost, elasticity | 3.31   |                                     |
| $\phi$      | Matching function, scale    | 0.19   | Patent traded share                 |

Table 4: Parameter Values from the Minimum Distance Estimation

| Targets                                         | Data  | Model |
|-------------------------------------------------|-------|-------|
| Economic growth rate(1981-1985)                 | 3.05% | 3.05% |
| R&D cost/sales of H firms(1981-1985)            | 3.62% | 3.63% |
| R&D cost/sales of L firms(1981-1985)            | 2.83% | 2.83% |
| Avg. number of industries of H firms(1981-1985) | 11.81 | 11.81 |
| Avg. number of industries of L firms(1981-1985) | 1.92  | 1.92  |
| The share of patents transacted(1981-1985)      | 30.9% | 30.9% |

Table 5: Model Fit for Key Moments in the Initial balanced growth path

# Recalibration - Post-Reform Balanced Growth Path

Set  $\sigma = 0.26$

Re-estimate parameters targeting 1996-2000

|             | Old Values | New Values | Interpretation                     |
|-------------|------------|------------|------------------------------------|
| $\phi$      | 0.19       | 0.28       | Matching efficiency increase       |
| $\theta$    | 0.16       | 0.20       | Sellers' bargaining power increase |
| $\mu$       | 1.5E-4     | 2.7E-4     | Higher costs of large scope        |
| $1 + \iota$ | 3.31       | 3.56       | More DRS to scope                  |
| $\gamma$    | 1.72       | 1.53       | Goods ideas are more rare          |

Table 7: Changes of Parameter Values

| Targets                                         | Data  | Model |
|-------------------------------------------------|-------|-------|
| Economic growth rate(1996-2000)                 | 3.34% | 3.34% |
| R&D cost/sales of H firms(1996-2000)            | 3.15% | 2.11% |
| R&D cost/sales of L firms(1996-2000)            | 6.71% | 6.67% |
| Avg. number of industries of H firms(1996-2000) | 6.31  | 6.31  |
| Avg. number of industries of L firms(1996-2000) | 1.61  | 1.61  |
| The share of patents transacted(1996-2000)      | 44.1% | 44.1% |

Table 6: Model Fit for Key Moments in Recalibration

# Decomposition

Gauge contribution of each explanation to the observed change in target moments.

- Change one or two parameters at a time

|                                  | Prod. Scope | $R\&D(H)$ | $R\&D(L)$ | Patent Trade | Growth |
|----------------------------------|-------------|-----------|-----------|--------------|--------|
| Data                             | -           | -         | +         | +            | +      |
| Patent Market ( $\phi, \theta$ ) | 30%         | 260%      | 49%       | 109%         | 233%   |
| Efficiency ( $\phi$ )            | 31%         | 251%      | 18%       | 118%         | 178%   |
| Bargaining Power ( $\theta$ )    | -6%         | 8%        | 27%       | -5%          | 33%    |
| Tax Credit ( $\sigma$ )          | 7%          | -256%     | 12%       | -8%          | 137%   |
| Production Cost ( $\mu, \iota$ ) | 71%         | 233%      | 19%       | 3%           | -155%  |
| Rare Good Ideas ( $\gamma$ )     | -9%         | 42%       | 4%        | -1%          | -134%  |

Table 8: Effects of Key Parameters

# Empirical Analysis

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Is there causality from the pro-patent reform to shrinkage of production scope of firms?

Exploit regional difference in exposure to policy changes.

- Establishment of Court of Appeals for the Federal Circuit. (CAFC) in 1982

# Empirical Strategy

Before CAFC, appeals of legal disputes involving patents were handled by regional circuits.

- Heterogeneous degree of patent protection across the country.

| Circuit Court | Invalidation rate |                 |
|---------------|-------------------|-----------------|
|               | Before            | After           |
| Boston        | 0.64              | 0.18            |
| New York      | 0.58              | 0.28            |
| Philadelphia  | 0.74              | 0.32            |
| Richmond      | 0.47              | 0.26            |
| New Orleans   | 0.36              | 0.20            |
| Cincinnati    | 0.60              | 0.30            |
| Chicago       | 0.54              | 0.30            |
| St.Louis      | 0.49              | 0.33            |
| San Francisco | 0.51              | 0.29            |
| Denver        | 0.27              | 0.22            |
| Atlanta       | 0.41              | 0.28            |
| DC            | 0.59              | — <sup>24</sup> |

Table 11: Patent Invalidation Rates in District Courts under Different Circuit Courts

## Prediction:

Higher invalidation rate before CAFC  $\Rightarrow$  Larger decrease in production scope.

$$\ln(ind_{ist}) = \alpha_i + \beta * inval_{c,pre} * post_t + \gamma X_{ist} + \mu_t + \varepsilon_{ist}$$

- $\ln(ind_{ist})$ : number of 6-digit NAICS industries of the firm  $i$  located in state  $s$ .
- $inval_{c,pre}$ : invalidation rate of circuit  $c$  that  $s$  belong to.
- $post$ : Year before or after CAFC.
- $X_{ist}$ : Control variables.

**Sample:** innovating firms in the LBD.

# Empirical Results

| Dependent Variable         | Ln(Number of Industries) |                      |                      |                      |
|----------------------------|--------------------------|----------------------|----------------------|----------------------|
|                            | (1)                      | (2)                  | (3)                  | (4)                  |
| Invalidation Rate*Post     | -0.0326**<br>(0.014)     | -0.0326**<br>(0.014) | -0.0332**<br>(0.014) | -0.0281**<br>(0.013) |
| Ln(Employment)             | 0.0888***<br>(0.007)     | 0.0899***<br>(0.007) | 0.0893***<br>(0.007) | 0.0894***<br>(0.007) |
| Post                       | 0.0196**<br>(0.008)      | 0.0218***<br>(0.007) |                      |                      |
| Ln(Real GDP)               |                          | -0.00487<br>(0.012)  |                      | -0.0332**<br>(0.014) |
| Effective Federal Tax Rate |                          | 0.272***<br>(0.036)  |                      | -1.462***<br>(0.508) |
| Effective State Tax Rate   |                          | 0.234*<br>(0.132)    |                      | -0.689**<br>(0.278)  |
| Federal R&D Tax Credits    |                          | 0.532***<br>(0.137)  |                      | -10.27**<br>(3.697)  |
| State R&D Tax Credits      |                          | -0.068<br>(0.141)    |                      | -0.0807<br>(0.127)   |
| Year-fixed Effects         | NO                       | NO                   | YES                  | YES                  |
| Firm-fixed Effects         | YES                      | YES                  | YES                  | YES                  |
| Observations               | 268000                   | 268000               | 268000               | 268000               |
| R-squared                  | 0.944                    | 0.944                | 0.944                | 0.944                |

Standard errors are clustered in the circuit court region by the post dummy level, shown in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

# Conclusion

Novel pattern of firm specialization (1980s and 1990s):

- Firms narrowed down production scope.
- Small firms increased R&D, Large firms decreased it.

Hypothesis: Tradability of intellectual property.

An endogenous growth model examines the roles of the new hypothesis:

- Accounts for 30% of scope decrease.
- Accounts for 49% of reallocation of innovation activities.
- Increase growth rate in 0.68%

Provide empirical support for causality from patent protection to contraction in scope.

- Uses regional and firm-level differences in the exposure to patent policy changes.